

LaosSA: An Approach Based on Fine-tuning Pre-trained Language Models for Lao Sentiment Analysis

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Abstract

Effective sentiment analysis for the Lao language is hindered by a scarcity of annotated resources and the inherent noise of real-world user feedback, which often features code-mixing and informal script patterns. This study addresses these challenges by introducing a high-quality, filtered benchmark of 25,139 Lao-language reviews collected from popular digital applications. We evaluate classical baselines, a lightweight textual convolutional neural network model trained from scratch, and several state-of-the-art pre-trained language models, including XLM-RoBERTa, multilingual BERT, and a Lao-specific multilingual BERT, across full fine-tuning, low-rank adaptation, and k-fold cross-validation settings as applicable. Our experimental results demonstrate that XLM-RoBERTa achieves superior performance with 96.64% accuracy and an F_1 -macro score of 0.9617. The textual convolutional neural network also provides a strong neural baseline with 94.03% F_1 -macro on the fixed split, placing it between the sparse baselines and the strongest pre-trained language model variants. We also find that while Lao-oriented pre-training provides significant benefits, low-rank adaptation offers a highly efficient performance-cost trade-off, maintaining competitive accuracy with minimal trainable parameters. Our research highlights the robustness of multilingual transfer for low-resource sentiment classification and provides a publicly available dataset and code at <https://github.com/KT246/LaosSA>.

Index Terms

Lao Sentiment Classification, Pre-trained Language Models, TextCNN, LoRA, Low-resource NLP.

1 Introduction

Sentiment analysis has emerged as a pivotal instrument in the modern digital economy, enabling service providers to distill actionable insights from the massive volumes of unstructured feedback generated across e-commerce and digital-service platforms [1]–[3]. For the Lao digital ecosystem, the automated classification of user sentiment represents an essential yet under-addressed frontier. Effective sentiment classification in Lao could facilitate scalable analysis of customer satisfaction across app stores and social media, yet the field remains constrained by a persistent scarcity of annotated corpora, standardized benchmarks, and mature Natural Language Processing (NLP) toolchains [4]–[6].

This resource disparity is further exacerbated by the intrinsic linguistic complexities of the Lao script. As a continuous, non-linear script (an abugida) lacking explicit word delimiters, Lao poses significant challenges for standard tokenization and morphological parsing [5]. Furthermore, real-world digital discourse in Lao is frequently characterized by high levels of semantic noise, colloquialisms, and code-mixed patterns [7]–[9], which complicates the direct transfer of models trained on formal or high-resource datasets. Consequently, methodologies optimized for high-resource languages often fail to capture the nuanced polarity of Lao mobile application reviews.

The limitations of traditional rule-based and classical machine learning approaches—which typically demand extensive feature engineering or lack contextual depth—have motivated the adoption of Pre-trained Language Models (PLMs). However, generic multilingual PLMs often underrepresent low-resource languages, leading to sub-optimal representational mappings and tokenization inefficiencies [4]. This motivates a rigorous comparative analysis between state-of-the-art multilingual PLMs (XLM-RoBERTa [10], mBERT [11]), Lao-specific models (mBERT-Lao [4]), and lightweight neural and sparse baselines. Our study investigates the hypothesis that while Lao-oriented pre-training and multilingual transfer provide superior performance, parameter-efficient adaptation strategies can maintain these gains at a fraction of the computational training cost.

In this work, we present *LaosSA*, a comprehensive benchmark for Lao sentiment classification based on a large-scale, filtered dataset of 25,139 mobile application reviews. Our primary contributions are summarized as follows:

- We introduce a high-quality, manually annotated Lao sentiment dataset specifically curated from mobile application reviews, capturing the complexities of real-world colloquial script and code-mixing patterns.
- We establish a robust benchmark across diverse modeling paradigms, evaluating the efficacy of cross-lingual transfer, monolingual pre-training, Low-Rank Adaptation (LoRA) [12], and Textual Convolutional Neural Network (TextCNN) from-scratch neural training.
- We perform a multi-dimensional analysis of predictive performance, error patterns, and training efficiency, providing actionable insights into the trade-offs between model quality and computational resource requirements for Lao NLP.

The remainder of this paper is organized as follows. Section 2 contextualizes our study within the broader landscape of low-resource sentiment analysis and PLM evolution. Section 3 details the LaosSA dataset construction, covering data collection, annotation, and the cleaning pipeline. Section 4 formalizes the classification task and presents the modeling framework and fine-tuning strategies. Section 5 reports our experimental configurations, results, and critical discussion. Finally, Section 6 concludes the paper and delineates future research directions.

2 Related Work

The evolution of Lao sentiment analysis has followed the broader trajectory of PLM development and low-resource adaptation. As illustrated in Fig.1, the progress spans from general multilingual baselines to dedicated Lao-specific benchmarks.

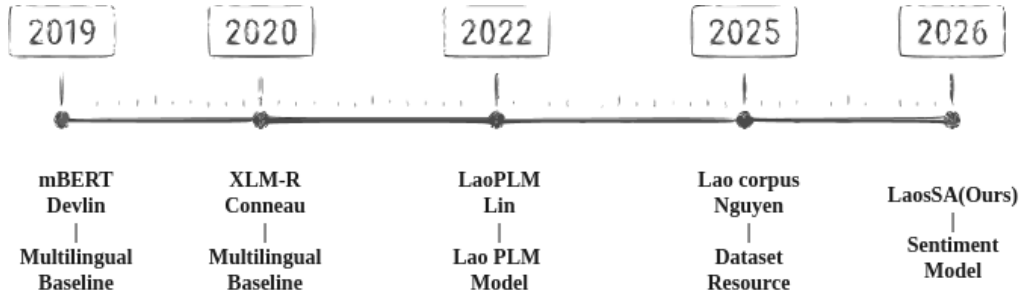


Fig. 1. Timeline of progress toward Lao sentiment analysis, showing the evolution from multilingual baselines and monolingual pre-trained models to the current LaosSA sentiment benchmark.

Early progress was marked by the introduction of multilingual BERT (mBERT) in 2019 [11], which provided the first massive transformer pre-training for over 100 languages, including zero-shot capabilities for low-resource scripts. However, a major limitation of mBERT remains its sub-optimal representation of Lao due to a shared vocabulary that assigns limited capacity to low-resource languages, often resulting in fragmented subword

tokens for continuous scripts. This was partially addressed by XLM-RoBERTa (XLM-R) in 2020 [10], which scaled up training data and refined cross-lingual representations. While XLM-R demonstrates significantly stronger zero-shot and few-shot performance, its primary limitation in the Lao context remains its generic pre-training objective, which does not explicitly account for the specific character boundaries or the noisy, informal code-mixing prevalent in modern digital communication.

A closely related regional example is Vietnamese e-commerce sentiment classification, where recent work reports strong results from fine-tuning PhoBERT, viBERT, and XLM-RoBERTa variants while also highlighting informal language, code-mixing, and domain-adaptation challenges that are relevant to our setting [3].

Low-resource language processing is still constrained by the lack of large corpora, annotated benchmarks, and mature preprocessing tools. For Lao, this limitation has been noted in work on word segmentation, text classification, and broader resource construction [4], [5]. Because Lao is written as a continuous script, preprocessing itself can become a source of error: tokenization and boundary detection are less straightforward than in whitespace-delimited languages [5]. This challenge is directly relevant to the current dataset, where many user reviews are very short and noisy.

The landscape shifted in 2022 with the introduction of LaoPLM [4], the first dedicated Lao-specific pre-trained model. Its strength lies in its ability to outperform generic multilingual models on formal Lao text classification by leveraging monolingual pre-training. Nevertheless, the evaluations in LaoPLM were constrained by relatively small datasets and formal registers, leaving the task of analyzing noisy, colloquial mobile app reviews largely unaddressed. More recently, in 2025, efforts by Nguyen et al. [5] established a semi-supervised word-segmented corpus using large language models. While this work successfully expanded the dataset resources for Lao, the focus remained on categorization and resource generation rather than on the nuanced polarity shifts found in short-text sentiments.

PLMs are now the dominant backbone for text classification because they provide contextualized representations that can be adapted across tasks [13], [14]. For Lao, the distinction between monolingual and multilingual PLMs is particularly important [4]. A Lao-oriented model can better reflect local tokenization behavior and lexical patterns, whereas multilingual models benefit from broader pretraining but may devote less capacity to low-resource languages [4], [13], [15].

Recent low-resource sentiment studies still report clear gains from transformer-based models, but they also show that performance depends on language fit, adaptation strategy, and dataset characteristics [7]–[9]. This is why our experiments compare XLM-RoBERTa and mBERT with mBERT-Lao, while also retaining a lightweight TextCNN and strong sparse baselines in the benchmark.

Our work (LaosSA, 2026) represents the next milestone by addressing these gaps across three dimensions: scale, domain, and adaptation strategy. Unlike prior benchmarks, LaosSA provides a large-scale (25,139 samples) filtered dataset specifically curated from mobile application reviews, capturing real-world noise and code-mixed patterns. Furthermore, we evaluate both full fine-tuning and Parameter-Efficient Fine-Tuning (PEFT) via LoRA to establish practical deployment baselines that were previously underexplored for the Lao digital ecosystem.

3 Dataset Construction and Analysis

This section details the construction and systematic refinement of the Lao sentiment benchmark, covering data collection from digital-service platforms, the manual annotation process, and the multi-stage filtering pipeline designed to ensure dataset quality.

3.1. Data Collection

The reviews used in this study were collected from Google Play applications that are widely used in Laos. The corpus includes comments from banking apps, social media apps,

and other digital-service applications rather than from a single e-commerce marketplace. This choice was practical: these applications produced enough Lao user feedback to build a usable sentiment dataset, and they reflect the short mobile reviews that a deployed Lao sentiment system would need to handle.

Each record contains two fields, namely the review text and a binary sentiment label. The final benchmark also retains short code-mixed reviews in which Lao words appear together with English interface terms such as *app*, *bug*, or *OK*, provided that the overall sentiment remains interpretable. This choice keeps the released benchmark closer to real mobile feedback in Laos and directly corresponds to the code-mixing setting referenced in the released resources.

3.2. Annotation Scheme

The raw reviews were manually annotated with binary sentiment labels. Each review was assigned a single label based on its overall polarity: label 1 for positive sentiment and label 0 for negative sentiment. The annotation process focused on the dominant sentiment expressed in each review, including explicit praise, complaints, dissatisfaction, or statements about overall user experience.

To improve label consistency, the annotation followed a simple set of practical guidelines. Reviews expressing clearly favorable opinions toward an application, service quality, or user experience were marked as positive, whereas reviews expressing frustration, malfunction, dissatisfaction, or criticism were marked as negative. During dataset refinement, noisy, invalid, or suspicious samples were revisited through the cleaning pipeline described below so that only reliable sentiment instances were retained in the final corpus.

3.3. Graphetics and Tonogenesis in Lao

Lao constitutes a tonal linguistic system characterized by an orthographic depth that departs fundamentally from the linear, segment-delimited paradigms of Latinate scripts. As an abugida, the Lao script employs a non-linear graphemic topology where vowels and tonemic diacritics are spatially distributed relative to a primary consonant nucleus. This configuration presents substantial obstacles for computational sentiment analysis, as the absence of explicit whitespace complicates morphological parsing and increases semantic entropy [16], [17]. The phonological foundation of Lao is anchored in the tri-partite classification of consonants-High, Mid, and Low registers-which serve as the phonetic anchors for tonogenesis. Furthermore, the positional variability of vowels, which may be preposed, postposed, superposed, or subposed, introduces significant encoding noise in informal digital text where characters often appear in visual rather than logical sequences.

Beyond synchronic graphetics, the tonogenetic evolution of Lao has produced a complex mapping where surface-level pitch contours emerge from the synergistic interaction of consonant class, syllable weight (live vs. dead), and explicit tone markers [18]. In the colloquial registers of user reviews, these markers are frequently omitted or substituted by phonological approximations, leading to considerable polarity ambiguity. Tables II and I provide a consolidated overview of the graphetic properties and tonogenetic constraints governing the Vientiane (standard) dialect, illustrating the dense linguistic logic that PLMs must reconstruct to achieve high-fidelity classification in this low-resource setting.

TABLE I
LAO LINGUISTIC FRAMEWORK: TONOGENETIC RULES (INTERACTIVE DETERMINATION).

Class	Live (No Mark)	Mai Ek (◌ᵉ)	Mai Tho (◌ᵒ)	Dead (Short)	Dead (Long)
High	Rising	Low Falling	High	High	Low Falling
Middle	Low Rising	Mid	High Falling	High	Low Falling
Low	Mid	High Rising	High Falling	Mid	High Falling

TABLE II
LAO LINGUISTIC FRAMEWORK: GRAPHETIC ARCHITECTONICS (CONSONANTS AND VOWELS).

Register	Consonant Graphemes	
High	ຂ, ສ, ຖ, ຜ, ຝ, ທ	
Middle	ກ, ຈ, ດ, ຕ, ບ, ປ, ຢ, ອ	
Low	ຄ, ງ, ຊ, ຍ, ນ, ພ, ຟ, ມ, ລ, ວ, ຮ	
Topology	Vowel Graphemes	Spatial Relation to Nucleus
Preposed	ເ, ຜ, ໂ, ິ, ໂ	Antecedent (Before)
Postposed	ຮ, າ, ັ	Subsequent (After)
Superposed	ື, ຸ, ຶ, ື, ັ	Superjacent (Above)
Subposed	ູ, ຶ	Subjacent (Below)

3.4. Data Preprocessing and Cleaning Pipeline

Before final training, the raw reviews were cleaned and filtered in several stages. After manual labeling, we removed noisy content such as repeated characters, emojis, invalid symbols, and strings that did not form meaningful Lao text. Particular attention was paid to Lao-specific character errors, especially cases in which vowels appeared without consonants or in which character combinations were visually present but linguistically invalid.

To assess label quality more efficiently, we trained a lightweight screening model and compared its predictions with the manual annotations. Samples that repeatedly agreed with their labels were retained, whereas mismatched cases were rechecked and many were removed. This step served as a practical filter for suspicious or low-quality reviews before the final benchmark experiments.

We also used a separate 5-fold cross-validation auditing step for difficult cases during dataset cleaning. This step was part of data filtering only and is distinct from the 3-fold cross-validation used later for model evaluation. Reviews that were repeatedly misclassified across folds were treated as unstable, and many of them contained invalid Lao character patterns or meaningless text fragments. Table III provides specific examples of the common script errors, lexical inconsistencies, and semantic noise identified through this systematic cleaning process, comparing noisy processed samples with their standard Lao corrections.

TABLE III
COMPARISON OF PREPROCESSING RESULTS BETWEEN PROCESSED DATA AND STANDARD LAO,
ILLUSTRATING COMMON LAO SCRIPT ERRORS IDENTIFIED DURING CLEANING.

Error Description	Processed Data (version 1)	Standard Lao	English Meaning
Vowel error & missing a letter: (1) ຢຸ → ຢູ: missing long-vowel sign + tone mark	ກະດິຢຸ (1)	ກະດິຢູ	it is good
Incorrect label & character error: (2) ບໍ → ບູ: missing tone mark; label mismatch	ບໍມີຫຍັງ (2)	ບູມີຫຍັງ	no problem
(3) ໃສ່ → ໃຊ້: wrong middle word; label mismatch	ພໍໃສ່ໄດ້ (3)	ພໍໃຊ້ໄດ້	enough to use
(4) ໄດ້ → ໄດ້: wrong leading vowel sign; label mismatch	ຕິດຕັ້ງບໍ່ໄດ້ (4)	ຕິດຕັ້ງບໍ່ໄດ້	cannot install
(5) ໄຈ → ໃຈ: wrong leading vowel sign; label mismatch	ຂອບໄຈ (5)	ຂອບໃຈ	thank you
Has no semantic meaning: (6) no valid Lao word sequence	ພາສາພ່າໄພຍາຍຈາຍ (6)		
(7) punctuation and noise fragment	. . ຕ. ວບໍ.ບຜຊໃ (7)		
(8) non-word string	ອອນສສືອ (8)		
(9) non-word string	ທາດດແພພະນ້າສສ (9)		

Note: Red text marks the incorrect span in Processed Data, while blue text marks the corrected span in Standard Lao. Teal markers (e.g., (1)) link error descriptions to specific occurrences in the text.

After this filtering process, the remaining reviews formed the final dataset used for training and validation.

3.5. Dataset Statistics

Table IV summarizes the final train and validation splits after cleaning and filtering. The two partitions remain closely matched in class balance and contain no missing text or labels. The reviews are short on average, with only a small number of much longer comments, which is consistent with the brief and noisy style of app-store feedback.

TABLE IV
FINAL TRAIN AND VALIDATION SPLIT STATISTICS FOR THE LAO REVIEW BENCHMARK AFTER FILTERING AND PREPROCESSING.

Statistic	Train	Validation
Rows	20,111	5,028
Columns	2	2
Empty text	0	0
Empty label	0	0
Unique text	20,095	5,028
Duplicated text within split	16	0
Negative reviews (Label = 0)	6,312 (31.39%)	1,578 (31.38%)
Positive reviews (Label = 1)	13,799 (68.61%)	3,450 (68.62%)
Minimum text length (characters)	2	3
Average text length (characters)	29.41	30.29
Maximum text length (characters)	618	506
Minimum word count	1	1
Average word count	1.52	1.52
Maximum word count	66	77

4 Methodology

This section presents the formal task definition for Lao sentiment classification, introduces the selected PLMs, and describes the fine-tuning strategies and training objectives employed in our modeling framework.

4.1. Problem Definition

We formulate Lao sentiment classification as a supervised binary text classification task over cleaned Lao reviews. Let the benchmark dataset be defined as

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N, \quad (1)$$

where each review $x_i = (w_1, w_2, \dots, w_n)$ is represented by the token sequence produced by a given tokenizer, and each label satisfies $y_i \in \{0, 1\}$, with 0 denoting negative sentiment and 1 denoting positive sentiment. Each review receives a single sentence-level label corresponding to its dominant polarity, even when the text remains short, informal, or noisy.

The task is to learn a classifier f_ψ that maps a Lao review to one of the two sentiment classes. For each input x , a model estimates a posterior distribution over the label space and predicts the dominant polarity according to

$$\hat{y} = f_\psi(x) = \arg \max_{c \in \{0, 1\}} p_\psi(y = c | x), \quad (2)$$

where ψ denotes the parameters of either a classical classifier or an adapted PLM-based classifier, and c indexes the candidate sentiment class. This formulation keeps the task definition model-agnostic so that sparse baselines and PLM-based systems can be compared under the same prediction rule.

The benchmark is defined under several practical assumptions. Each review is treated as an independent sample without conversational context or user metadata. The experiments focus on document- or sentence-level polarity classification rather than aspect-level sentiment extraction, and all systems operate directly on the cleaned review text rather than on hand-crafted linguistic features. These assumptions keep the comparison centered on representation quality and adaptation strategy rather than on feature design or external metadata.

The evaluation protocol follows the final filtered split described in Section 3.5. For the main benchmark, models are trained on 20,111 training samples and evaluated on a fixed validation set of 5,028 samples. In addition, a separate 3-fold cross-validation

setting is used as a robustness check on the training partition. Performance is reported using Accuracy, Precision-macro, Recall-macro, and F_1 -macro, with F_1 -macro treated as the primary comparison metric because of the moderate class imbalance between positive and negative reviews.

4.2. Pre-trained Language Models

We evaluate three PLMs: XLM-RoBERTa [10], mBERT [11], and mBERT-Lao [4]. XLM-RoBERTa and mBERT serve as multilingual references for cross-lingual transfer [13]–[15], while mBERT-Lao provides a Lao-supporting alternative [4]. Table V summarizes the main properties of the three models under the benchmark-facing names used throughout this paper. All PLM runs share the same preprocessing and evaluation protocol, and the LoRA setting attaches adapters to the query, key, and value projections.

TABLE V
PRE-TRAINING AND ARCHITECTURAL COMPARISON OF THE PLMS USED IN THE BENCHMARK.

Aspect	XLM-RoBERTa	mBERT	mBERT-Lao
Language scope	Multilingual (100 languages)	Multilingual (104 languages)	Lao-supporting
Pre-training data	CommonCrawl multilingual corpus (~2.5 TB)	Multilingual Wikipedia corpus	Lao-oriented RoBERTa-style pretraining
Tokenizer	SentencePiece, ~250k subwords	WordPiece, ~119k vocabulary	RoBERTa tokenizer, ~50k vocabulary
Architecture (base)	12 layers, 768 hidden, 12 heads	12 layers, 768 hidden, 12 heads	12 layers, 768 hidden, 12 heads
Role in this benchmark	Strong multilingual reference	Strong multilingual reference	Lao-supporting comparison model

In addition to the PLMs, the benchmark includes four non-PLM references: Logistic Regression, Support Vector Machine (SVM), Decision Tree [19], and TextCNN [20]. The first three use Term Frequency-Inverse Document Frequency (TF-IDF) character n -gram features, while TextCNN is evaluated under from-scratch training as a compact convolutional classifier with 128-dimensional embeddings, 128 filters per kernel size $\{3, 4, 5\}$, and dropout 0.3. This gives the benchmark a lightweight neural baseline that does not rely on pretrained language-model weights and allows for a direct comparison between transfer-learning and static initialization paradigms.

4.3. Fine-Tuning and From-Scratch Training Strategies

Each PLM is trained with either Full Fine-tuning or LoRA, while the neural baseline is trained from-scratch. Full Fine-tuning updates the backbone and classification head end to end. LoRA instead freezes the pretrained weights and learns low-rank updates in selected attention projections [12], [21], [22]. From-scratch training involves random weight initialization and localized optimization without knowledge transfer. We also report 3-fold cross-validation as a robustness check; in that setting, a separate model is trained on each fold and the final score is computed from the concatenated out-of-fold predictions.

Let θ denote the backbone parameters and ϕ the classification head. Full Fine-tuning solves

$$(\theta^*, \phi^*) = \arg \min_{\theta, \phi} \mathcal{L}_{\text{train}}(\theta, \phi). \quad (3)$$

For LoRA, the pretrained matrix is decomposed as $W = W_0 + BA$, where W_0 is frozen and $A \in \mathbb{R}^{r \times d}$ and $B \in \mathbb{R}^{d \times r}$ are trainable. The optimization becomes

$$(A^*, B^*, \phi^*) = \arg \min_{A, B, \phi} \mathcal{L}_{\text{train}}(W_0 + BA, \phi). \quad (4)$$

For cross-validation, out-of-fold predictions from all K folds are concatenated and scored jointly. The reported score is therefore

$$\text{Score}_{\text{CV}} = \text{Score} \left(\bigcup_{k=1}^K \hat{D}_{\text{val}}^{(k)} \right), \quad K = 3, \quad (5)$$

where $\hat{D}_{\text{val}}^{(k)}$ denotes the predictions produced by the best checkpoint of fold k on its held-out partition.

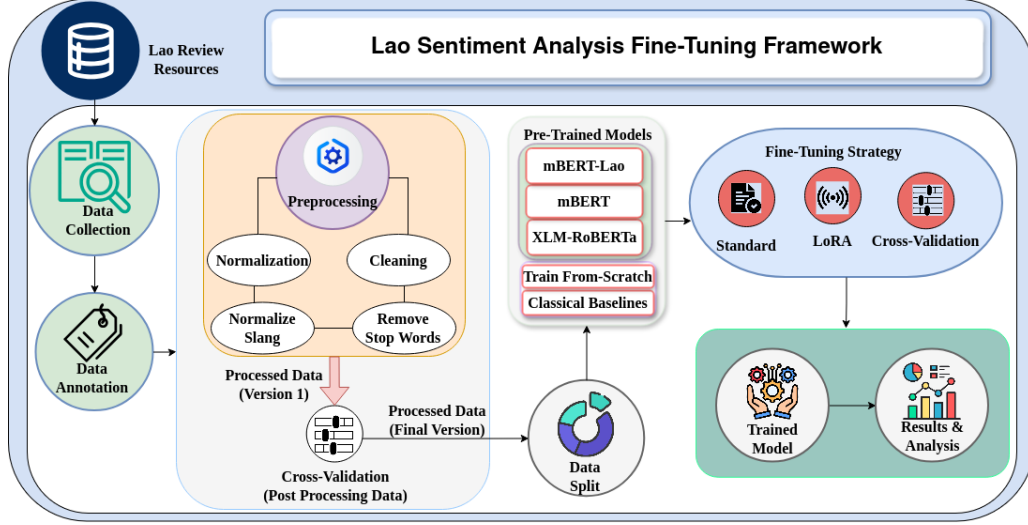


Fig. 2. Overall methodology framework for Lao sentiment classification, including dataset preparation, model selection, and training strategies.

Fig.2 summarizes the benchmark in three stages. The workflow begins with Google Play review collection, binary annotation, normalization, cleaning, and the post-processing audit used to construct the final split. It then trains XLM-RoBERTa, mBERT, mBERT-Lao, TextCNN, and TF-IDF baselines under the strategy variants applicable to each family, including Full Fine-tuning, LoRA, from-scratch training, and 3-fold cross-validation. Finally, the pipeline exports checkpoints, validation predictions, standard metrics, timing summaries, and false-positive/false-negative artifacts for the later analysis sections.

4.4. Training Objective

PLM runs minimize weighted cross-entropy loss. Class weights compensate for the moderate label imbalance in the final dataset. For a training set of N samples and two sentiment classes, the objective is

$$\mathcal{L} = - \sum_{i=1}^N \sum_{c=1}^2 w_c y_{i,c} \log \hat{y}_{i,c}, \quad (6)$$

where w_c is the weight for class c , $y_{i,c}$ is the one-hot target, and $\hat{y}_{i,c}$ is the predicted probability for class c .

We train the PLMs with standard pretrained-model tooling and fine-tuning practice [13], [14], using a learning rate of $1e^{-5}$, weight decay 0.01, warmup ratio 0.1, and maximum sequence length 128. For Full Fine-tuning and LoRA, the best checkpoint is selected by validation loss. TextCNN uses the same epoch budget, batch sizes, and sequence length, but it initializes all parameters from scratch. The classical baselines use the same split and train TF-IDF character n -gram features from 3 to 5 with `min_df=2` and at most 50,000 features before classifier fitting.

Algorithm 1 summarizes the unified benchmark pipeline used throughout the experiments. It reflects the fixed-split baseline runs, the PLM-based fixed-split runs, and the separate 3-fold cross-validation setting reported later in Section 5.

Algorithm 1 Unified training and evaluation pipeline for Lao sentiment classification

Require: fixed-split $D_{\text{train}}, D_{\text{val}}$, PLM set $\mathcal{M}$, baseline set $\mathcal{B}$, neural reference set $\mathcal{N}$, strategies $\mathcal{S} = \{\text{FT}, \text{LoRA}\}$

Ensure: prediction files, metrics, timing statistics, and error-analysis artifacts

```

1: for all  $b \in \mathcal{B}$  do                                     ▷ Classical sparse baselines
2:   compute TF-IDF features  $\Phi(x)$  for  $x \in D_{\text{train}}$ 
3:    $\psi_b \leftarrow \arg \min_{\psi} \mathcal{L}_{\text{train}}(\psi, \Phi(D_{\text{train}}))$ 
4:    $\hat{y} \leftarrow f_{\psi_b}(\Phi(x))$  for  $x \in D_{\text{val}}$ , evaluate metrics
5: end for
6: for all  $n \in \mathcal{N}$  do                                     ▷ Neural models (from-scratch)
7:   randomly initialize  $\psi_n$ , minimize  $\mathcal{L}$  on  $D_{\text{train}}$  via Eq. (6)
8:    $\psi^* = \arg \min_{\psi} \mathcal{L}(\psi, D_{\text{val}})$ 
9:    $\hat{y} = f_{\psi^*}(x)$  for  $x \in D_{\text{val}}$ , store metrics
10: end for
11: for all  $m \in \mathcal{M}$  do                                     ▷ PLMs (fine-tuning)
12:   for all  $s \in \mathcal{S}$  do
13:     load tokenizer and pretrained checkpoint for  $m$ 
14:     if  $s = \text{LoRA}$  then
15:       inject LoRA adapters  $BA$  via Eq. (4)
16:     else
17:       enable end-to-end updates via Eq. (3)
18:     end if
19:     minimize  $\mathcal{L}$  on  $D_{\text{train}}$  via Eq. (6)
20:      $\psi^* = \arg \min_{\psi} \mathcal{L}(\psi, D_{\text{val}})$ 
21:     compute  $\hat{y} = f_{\psi^*}(x)$  for  $x \in D_{\text{val}}$ , store metrics
22:   end for
23:   for  $k = 1$  to 3 do
24:     partition  $D_{\text{train}} \rightarrow \{D_{\text{train}}^{(k)}, D_{\text{val}}^{(k)}\}$ 
25:      $\psi^{(k)} \leftarrow \text{optimize}(\mathcal{L}, D_{\text{train}}^{(k)}, m)$ 
26:      $\hat{D}_{\text{val}}^{(k)} \leftarrow \{f_{\psi^{(k)}}(x) \mid x \in D_{\text{val}}^{(k)}\}$ 
27:   end for
28:   compute  $\text{Score}_{\text{CV}}$  using Eq. (5)
29: end for
30: aggregate  $\hat{D}_{\text{val}}$  and metrics to produce final artifacts

```

5 Experiments

This section outlines the experimental configuration, baseline selections, and evaluation metrics. We then present a comparative analysis of our findings regarding predictive performance, error patterns, and computational efficiency across the evaluated models.

5.1. Experimental Setup

We begin by specifying the compared systems, the evaluation metrics, and the implementation details required to reproduce the benchmark.

5.1.1. Compared Models

The benchmark combines classical baselines, a lightweight neural baseline, and PLM-based models. The classical group contains Logistic Regression, SVM, and Decision Tree [19]. The neural baseline is TextCNN [20], evaluated under from-scratch training and 3-Fold Cross-Validation. The PLM group contains XLM-RoBERTa [10], mBERT [11], and mBERT-Lao [4], each evaluated with Full Fine-tuning, LoRA [12], and 3-Fold Cross-Validation. This setup creates a controlled comparison between sparse models, a compact neural model, and pretrained transformers on the same Lao review data.

The classical baselines were trained on the same fixed train-validation split used for the fixed-split PLM runs. They use character-level TF-IDF features with an n -gram range of 3-5, `min_df=2`, and at most 50,000 features. Logistic Regression and SVM were implemented through `SGDClassifier` with `log_loss` and `hinge` objectives, respectively,

while Decision Tree used `max_depth=40` and `min_samples_leaf=2`. TextCNN uses 128-dimensional trainable embeddings, 128 filters for each kernel size in $\{3, 4, 5\}$, and dropout 0.3, giving the benchmark a compact neural reference that does not depend on pretrained weights.

5.1.2. Evaluation Metrics

We report Accuracy, Precision-macro, Recall-macro, and F_1 -macro, which are standard evaluation measures for text classification and information retrieval tasks [1], [23]. Because the dataset is moderately imbalanced, F_1 -macro is treated as the main comparison metric, while the remaining metrics are used to show overall correctness and class-level balance.

For binary sentiment classification, Accuracy is defined as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \quad (7)$$

For each class $c \in \{0, 1\}$, let

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c}. \quad (8)$$

The macro-averaged scores reported in this paper are then

$$\text{Precision}_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C P_c, \quad \text{Recall}_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C R_c, \quad (F_1)_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C \frac{2P_c R_c}{P_c + R_c}, \quad (9)$$

where c indexes a sentiment class and $C = 2$ for the negative and positive labels. Here, TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively, while TP_c , FP_c , and FN_c are the corresponding class-specific counts used to compute P_c and R_c for class c .

5.1.3. Implementation Details

The experiments use the final filtered Lao review dataset described in Section 3.5. Logistic Regression, SVM, Decision Tree, and fixed-split TextCNN were trained on the same validation split as the fixed-split PLM runs, while 3-fold cross-validation was applied on the training partition for the PLMs and for TextCNN. The exact hyperparameters are listed in Table VI; the main difference across PLM runs is whether the full backbone is updated or only LoRA adapters are trained.

TABLE VI
TRAINING CONFIGURATION USED IN THE EXPERIMENTS.

Component	Configuration
PLM epochs	25
Batch size / eval batch size	16 / 32
Maximum sequence length	128
Learning rate	1×10^{-5}
Weight decay	0.01
Warmup ratio	0.1
Cross-validation	Stratified $K = 3$
LoRA setting	$r = 16$, $\alpha = 32$, dropout = 0.1
TextCNN setting	Embedding 128, filters 128, kernels 3/4/5, dropout 0.3
Best-model criterion	Validation loss
Baseline features	TF-IDF char n -grams (3-5), max 50,000

TABLE VII
RUNTIME ENVIRONMENT RECORDED IN THE EXPERIMENT ARTIFACTS.

Hardware item	Recorded value
CPU	1 physical core / 2 total cores
RAM	12.67 GB
GPU	Tesla T4
GPU memory	14.56 GB
Recorded hardware scope	14 experiment folders

5.2. Results and Discussion

We now examine the main performance pattern and use error and efficiency analyses to explain the observed trade-offs.

5.2.1. Overall Performance

Taken together, Table VIII and Fig.3 indicate that backbone suitability matters more than simply moving from sparse to neural baselines, or from LoRA to Full Fine-tuning within the same PLM family. TextCNN provides a strong middle point: on the fixed split it outperforms Logistic Regression and approaches SVM, yet the best scores still come from the XLM-RoBERTa variants. This interpretation is consistent with low-resource sentiment studies that attribute transfer quality mainly to language fit and data characteristics rather than to simply updating more parameters [7], [8], [24]. The benchmark also shows that sparse baselines remain meaningful for Lao app reviews, suggesting that the task still contains strong lexical sentiment cues even when pretrained encoders are available.

TABLE VIII
OVERALL PERFORMANCE GROUPED BY MODEL FAMILY AND TRAINING STRATEGY.

Model Family	Training Strategy	Acc.	F_1	Prec.	Rec.
XLM-RoBERTa	Full Fine-tuning	0.9664	0.9617	0.9544	0.9702
	LoRA	0.9642	0.9589	0.9539	0.9645
	Cross-Validation	0.9648	0.9597	0.9541	0.9659
mBERT	Full Fine-tuning	0.6969	0.6057	0.6336	0.6012
	LoRA	0.6901	0.6048	0.6253	0.6004
	Cross-Validation	0.6967	0.6137	0.6350	0.6084
mBERT-Lao	Full Fine-tuning	0.9407	0.9328	0.9241	0.9437
	LoRA	0.9326	0.9235	0.9154	0.9335
	Cross-Validation	0.9375	0.9285	0.9228	0.9350
TextCNN	From Scratch	0.9477	0.9403	0.9338	0.9478
	Cross-Validation	0.9339	0.9242	0.9192	0.9298
Classical baselines	Logistic Regression	0.9451	0.9370	0.9322	0.9425
	SVM	0.9505	0.9436	0.9363	0.9522
	Decision Tree	0.8878	0.8721	0.8662	0.8792

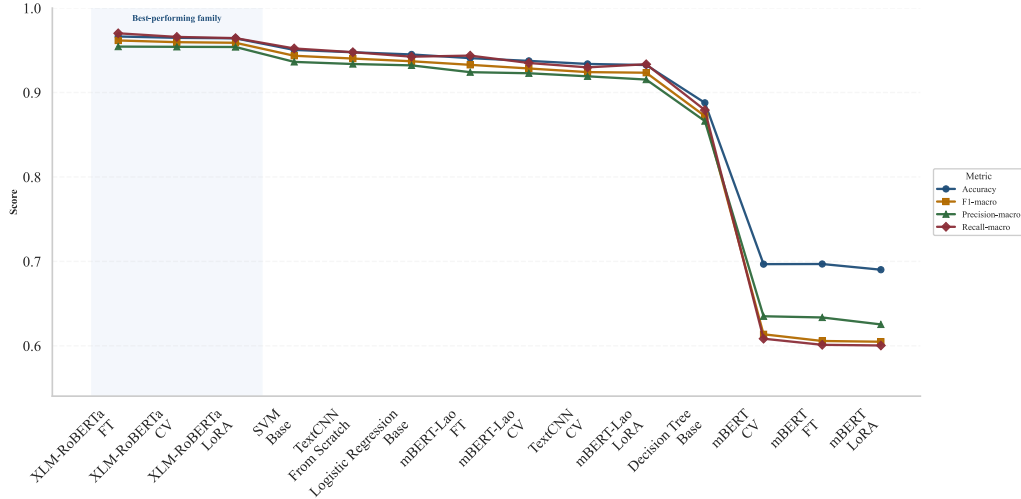


Fig. 3. Metric comparison across all benchmarked settings, sorted by F_1 -macro.

5.2.2. Domain Adaptation Analysis

The gap between mBERT-Lao and generic mBERT suggests that Lao support is not a minor implementation detail but a substantive source of performance difference. This interpretation aligns with LaoPLM and related low-resource work, which argue that multilingual pretraining can under-serve Lao when tokenization behavior and corpus coverage are mismatched [4], [24]. TextCNN further demonstrates that a compact neural model can learn substantive review-local patterns through from-scratch training, achieving competitive performance without the necessity of expensive pre-training stages. However, its slightly lower cross-validation scores relative to its fixed-split peak suggest that from-scratch neural learning remains more sensitive to distribution shifts and training-data variability than the robust representations provided by the strongest PLMs. At the same time, the competitiveness of SVM and Logistic Regression indicates that many app-review decisions are driven by local lexical cues and short formulaic complaints. In other words, domain adaptation in this dataset appears to depend on both language-aware representations and sensitivity to concise, review-specific expressions.

5.2.3. Error Analysis

Fig. 4 helps explain why the model ranking separates so clearly. The strongest systems maintain a more balanced error profile, whereas weaker multilingual transfer is associated with a clear positive-class bias. This matters because the difficult cases are not random; they are concentrated in short complaint-like comments, code-mixed fragments, and mixed-polarity reviews that combine a problem report with a request, update, or partially positive tone. Related low-resource sentiment studies likewise report that short, informal, and mixed-language comments are a recurring source of ambiguity [7], [8], but the present results suggest that these cases are especially consequential for Lao app reviews, where limited context and informal phrasing amplify ambiguity.

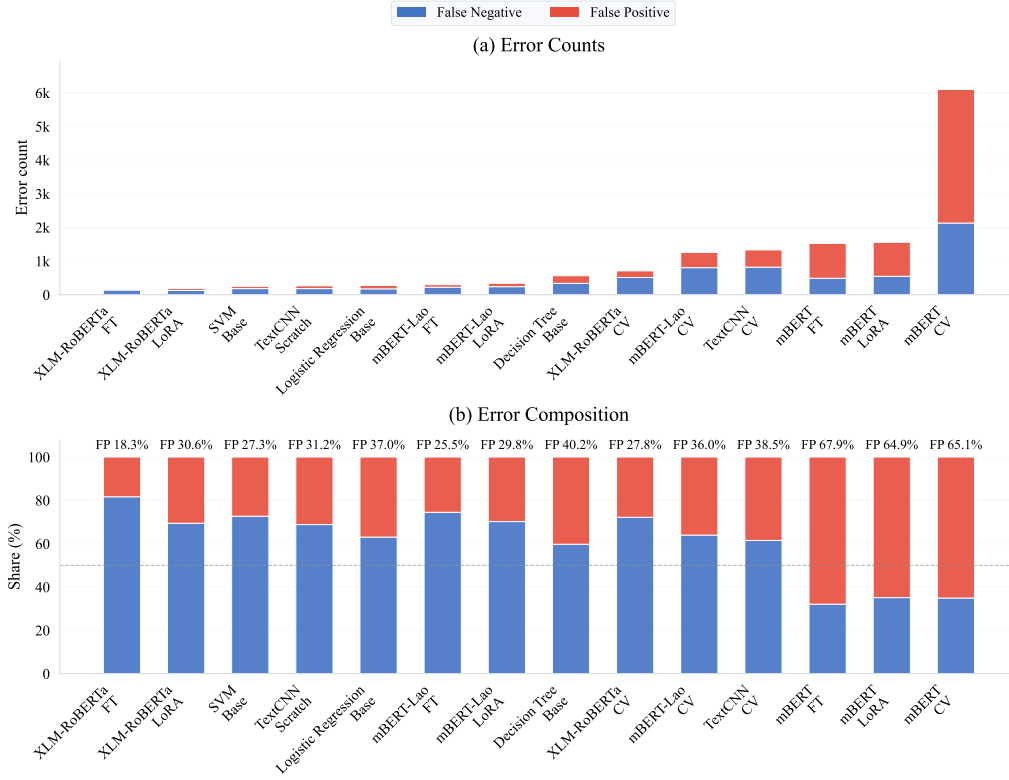


Fig. 4. Error-analysis overview based on the aggregated false-positive and false-negative samples from all experiments. Panel (a) reports raw false-negative and false-positive counts, while panel (b) shows the corresponding error composition percentages for the same experiments.

Table IX illustrates this pattern qualitatively. False positives are usually short complaints whose negative intent is expressed indirectly or through code-mixed wording, whereas false negatives more often contain softened criticism, requests, or mixed cues. The dominant failure mode is therefore semantic ambiguity in short feedback, rather than annotation instability alone.

TABLE IX
REPRESENTATIVE EXAMPLES OF FIXED-SPLIT ERROR PATTERNS

False Positive		False Negative	
Example	Label	Example	Label
ບໍ່ ok not OK	0	ເປັນຍັງຄື ທັງຫມົດທີ່ຈຶ່ງບໍ່ໄດ້ why does nothing work?	1
ປ່ຽນໂທລະສັບໃຫມ່ changed to a new phone	0	ດີກຢູ່ ແຕ່ໃຫ້ອັບເດດດູໂພດ it is good, but please update it	1
ເບີລະຫັດ whatsapp WhatsApp verification code	0	ແຕ່ໃຫ້ປັບປຸງໃຫ້ໄວກວ່າເກົ່າ please make it faster	1
bug ຫລາຍ too many bugs	0	ໂຕໂດດີ ຖ້າບໍ່ອັບເດດແອປ ແຮງຊີດີ the code is good, but the update hurts the app	1
ແອັບຊ້າສຸດ the app is very slow	0	ຂອບເຂດສົ່ງ ກ້ວາງໄກລ ສົ່ງໄວ ບໍ່ມີບັນຫາ wide service area, fast delivery, no problems	1

Fig.5 provides a complementary phrase-level view of the same error set using the

30 most frequent short false-positive and false-negative samples. This visualization is intended as a qualitative summary of recurring failure expressions rather than as a replacement for the quantitative counts in Fig.4.

Fig. 5. Phrase-level word clouds built from the 30 most frequent short false-positive and false-negative samples in the fixed-split experiments. Larger phrases indicate more frequently repeated failure cases.

TABLE X
EFFICIENCY COMPARISON BETWEEN FULL FINE-TUNING AND LoRA.

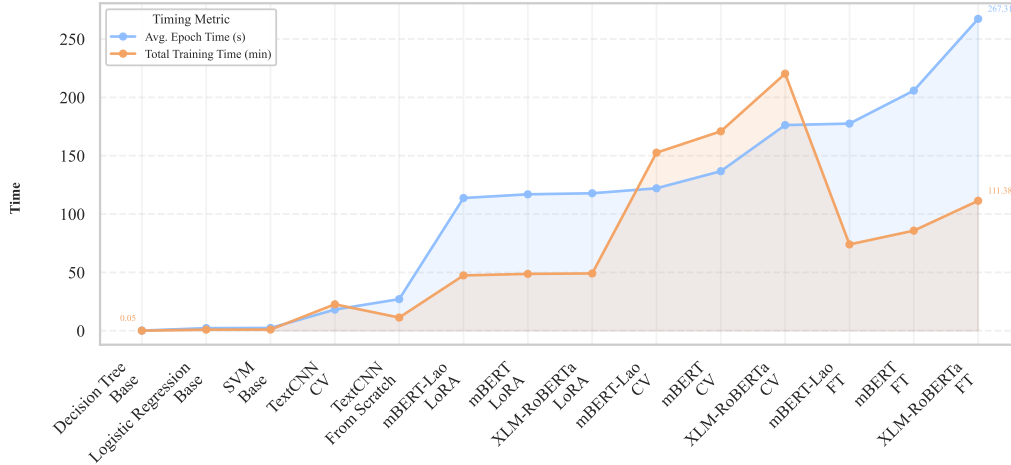


Fig. 6. Timing comparison across all benchmarked settings, sorted by average epoch time.

6 Conclusion

In this study, we established a comprehensive benchmark for Lao sentiment classification, utilizing a large-scale, filtered dataset of real-world mobile application reviews. Our systematic evaluation across diverse modeling paradigms-spanning classical sparse baselines, from-scratch neural references, and state-of-the-art PLMs-reveals that while XLM-RoBERTa delivers the most robust performance, the choice of adaptation strategy and language support are decisive factors for Lao NLP. Specifically, we demonstrate that (i) monolingual pre-training and cross-lingual transfer (XLM-RoBERTa) consistently outperform generic multilingual alternatives, (ii) PEFT via LoRA maintains competitive quality with minimal computational overhead, and (iii) lightweight from-scratch neural training (TextCNN) provides a credible reference that can learn review-specific patterns without relying on PLM weights.

Despite these advancements, the study identifies critical challenges posed by linguistic ambiguity, noise, and semantic shifts in code-mixed environments. Future research should prioritize enhancing dataset diversity and label consistency, while exploring more granular analysis frameworks, such as aspect-level sentiment classification, to provide deeper insights into user experience. By releasing our dataset and trained models, we aim to support the development of more scalable and accurate NLP tools for the Lao digital ecosystem.

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